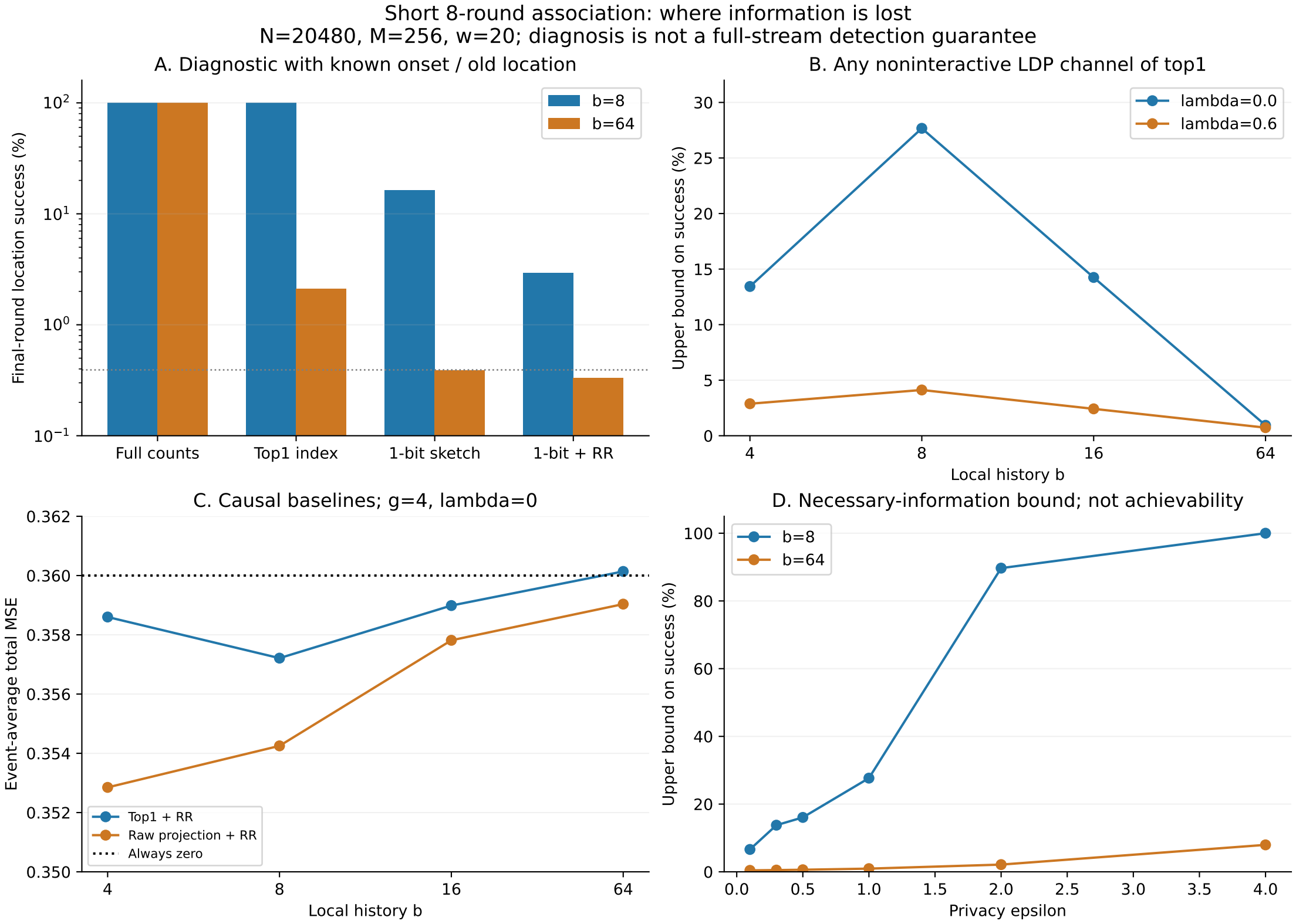
Short 8-round association: where information is lost
N=20480, M=256, w=20; diagnosis is not a full-stream detection guarantee
A. Diagnostic with known onset / old location
Final-round location success (%)
b=8
b=64
Full counts
Top1 index
1-bit sketch
1-bit + RR
B. Any noninteractive LDP channel of top1
Upper bound on success (%)
lambda=0.0
lambda=0.6
Local history b
C. Causal baselines; g=4, lambda=0
Event-average total MSE
Top1 + RR
Raw projection + RR
Always zero
Local history b
D. Necessary-information bound; not achievability
Upper bound on success (%)
b=8
b=64
Privacy epsilon